

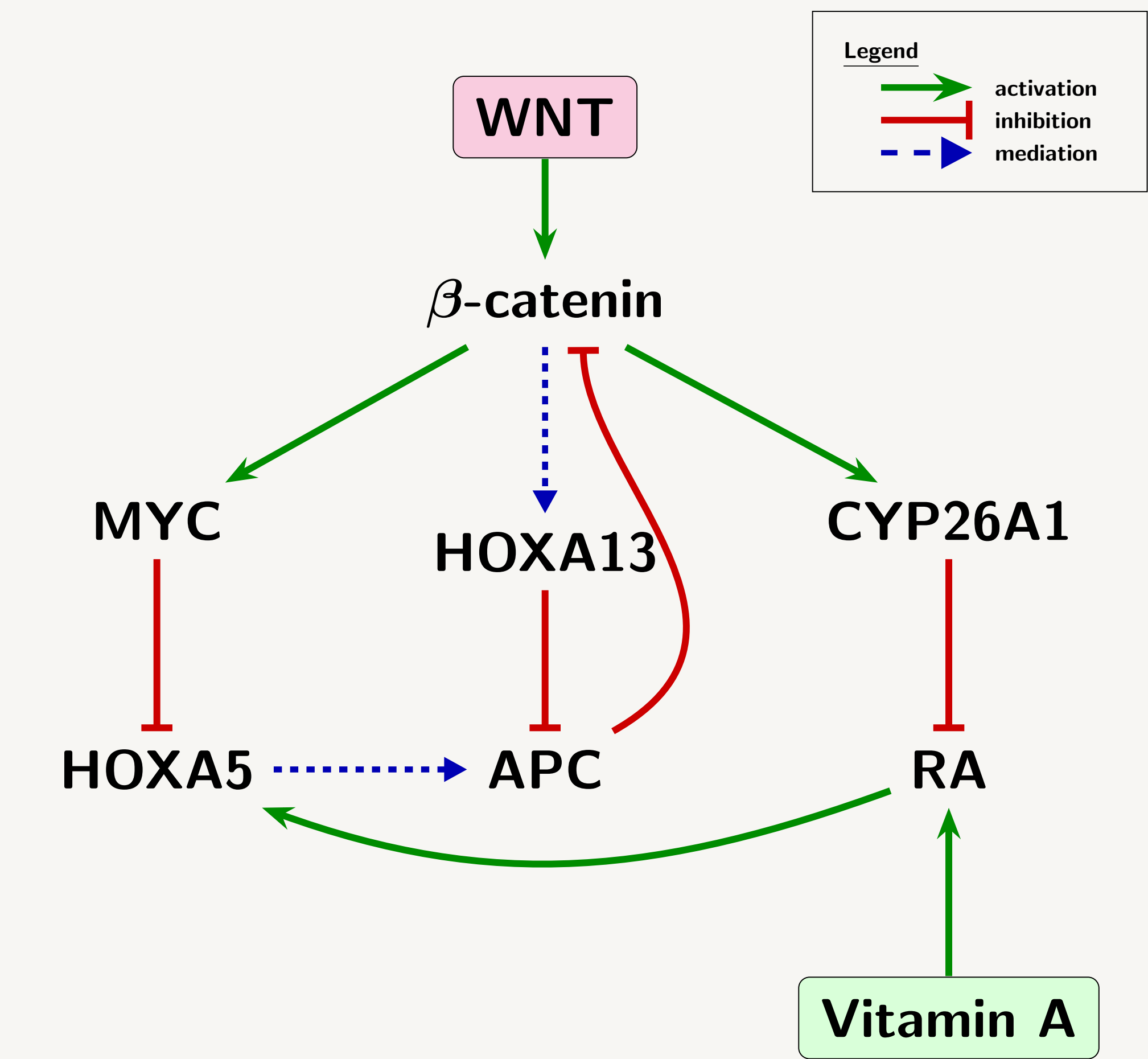
Physics-Informed Neural Networks and Neural-Mechanistic Hybrids for a WNT-RA-HOX Model of Colorectal Cancer Stemness

A nondimensionalized 7-ODE model, sparse-data forward solves, Bayesian identifiability, and learned terms.

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1 Seven equations, 36 parameters.

WNT signalling drives proliferation in the intestinal crypt, retinoic acid drives differentiation, and the HOX genes arbitrate. Loss of APC breaks that balance. We nondimensionalize a 7-ODE model of this network, solve it with a Fourier-feature PINN from 40 observations, invert it for the 36 parameters, and read the non-identifiability that remains off a Bayesian posterior. Replacing one mechanistic term with a neural network returns the problem in a sharper form: the constraint meant to protect the host equation is imposed where the data never reach. One knockdown condition, at a fixed budget, removes the error.



Green activates, red inhibits, dashed mediates. Seven species, one constitutive WNT drive, one vitamin-A input.

Nondimensionalization

Scale each species by its initial value and time by the β -catenin turnover rate $d_b = 1 \text{ hr}^{-1}$:

$$B = B_0 b, \quad P = P_0 p, \quad \dots, \quad \tau = d_B t$$

Every loss term normalizes to $-x$, so each equation collapses to one form with a pure timescale ratio in front:

$$\varepsilon_X \frac{dx}{d\tau} = \text{production} - x$$

For β -catenin, in full:

$$\frac{db}{d\tau} = W + \eta_{13} \frac{h_{13}^n}{\kappa_{13} + h_{13}^n} - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b}$$

The units and the count change, not the model. The system is stiff: $\varepsilon_R = 0.40$, $\varepsilon_M = 0.60$.

The readout: the stemness index

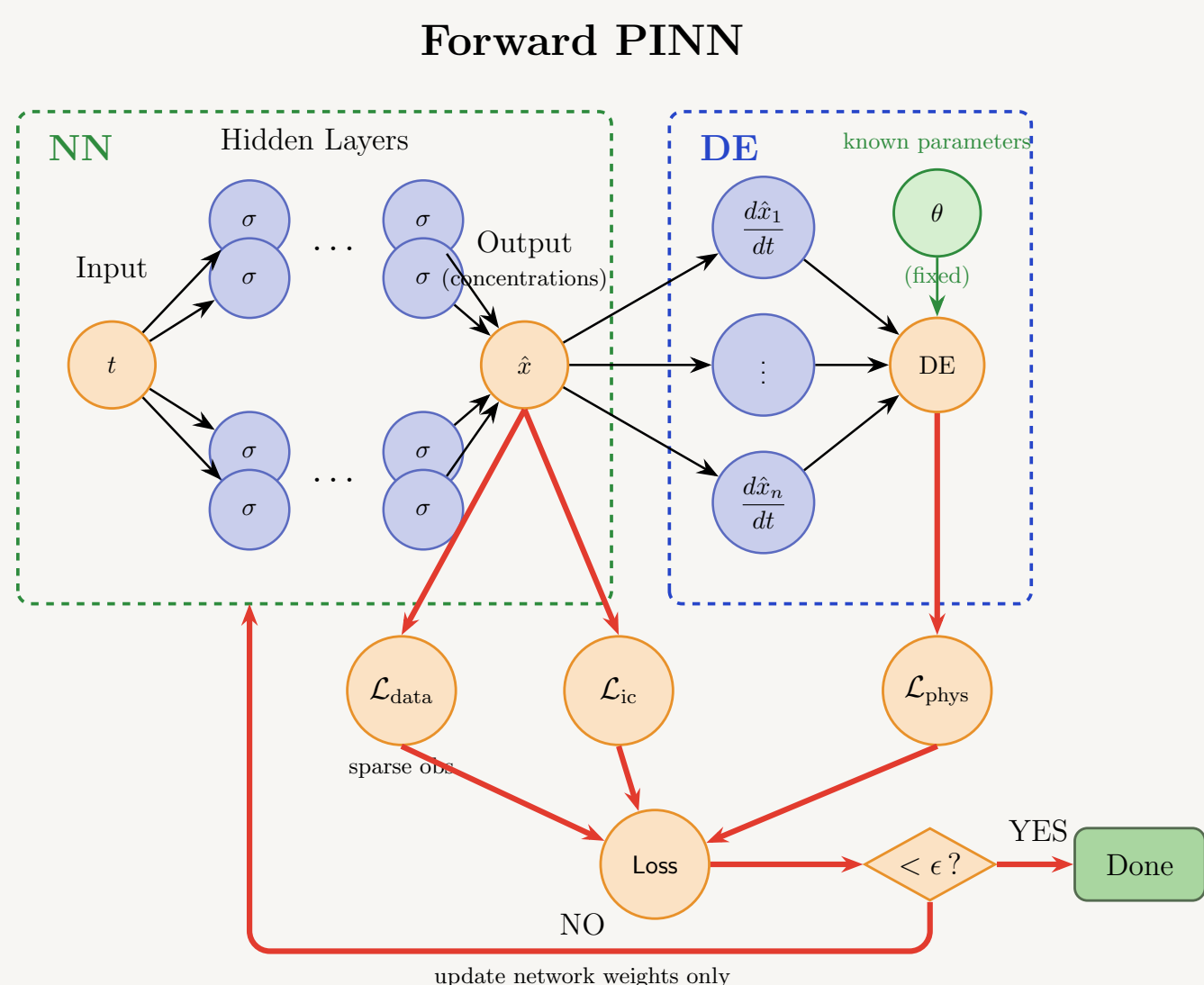
$$S(\tau) = \frac{b(1+\alpha_{13}h_{13})}{(1+p)(1+\alpha_5h_5)}$$

Four regimes, Normal to Severe APC Loss, differ only in the WNT drive W ($0.8 \rightarrow 2.0$) and APC functionality θ_P ($1.0 \rightarrow 0.25$).

Key references

M. Raissi, P. Perdikaris, G. E. Karniadakis. Physics-informed neural networks. J. Comput. Phys. **378**, 686 (2019).
C. Rackauckas et al. Universal differential equations for scientific machine learning. arXiv:2001.04385 (2020).

2 One network solves it, then inverts it.

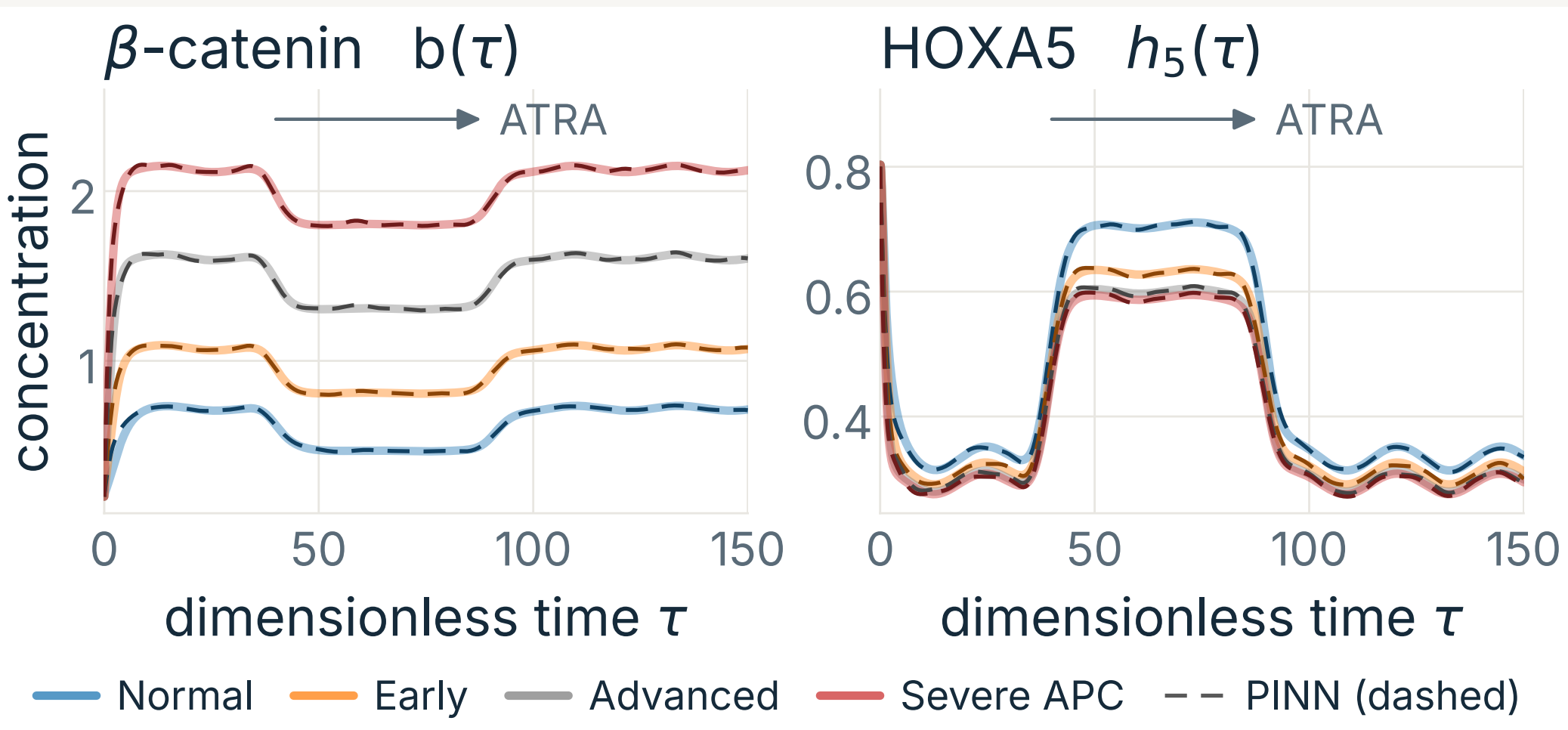


The network proposes a trajectory, and automatic differentiation feeds it back through the ODEs. The inverse network is the same graph with θ trainable.

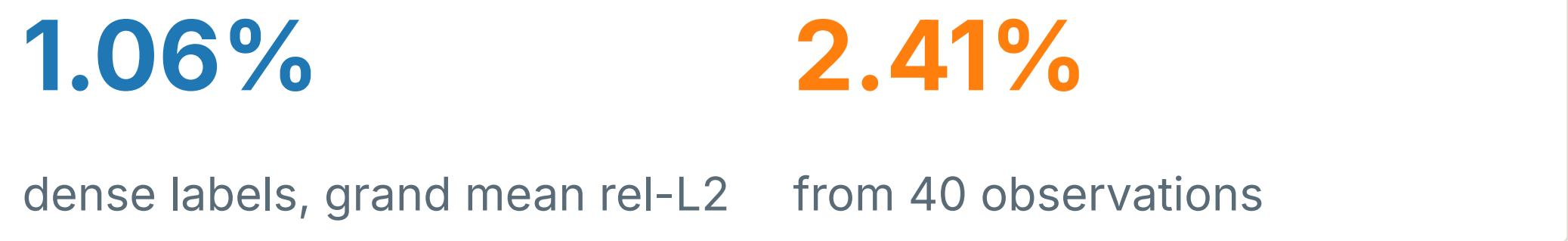
$$\mathcal{L} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{phys}} + 20\mathcal{L}_{\text{ic}}$$

Time enters through a fixed random Fourier embedding, 16 modes at $\sigma = 4$, then a 256-wide, 4-deep GELU network:

$$\gamma(\tau) = \left[\frac{\tau}{T}, \sin(2\pi \frac{\tau}{T} B_k), \cos(2\pi \frac{\tau}{T} B_k) \right]_{k=1}^{16}$$



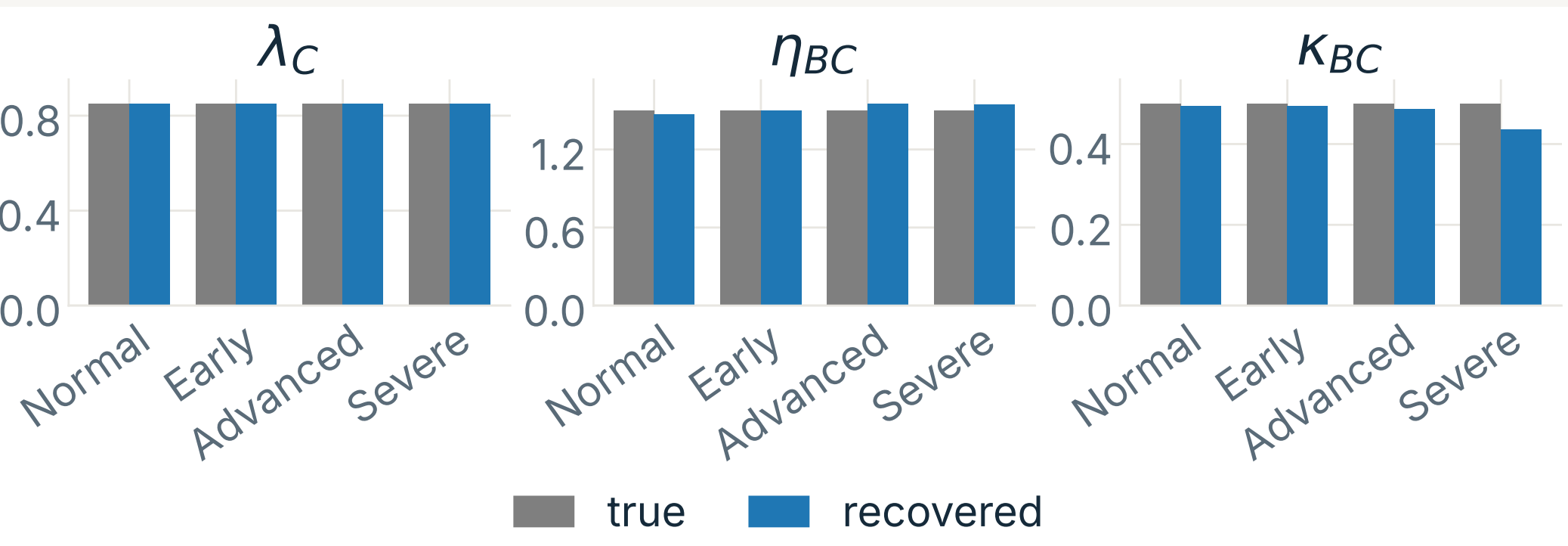
Trained on 40 sparse observations plus the physics residual, the PINN tracks the Radau reference in every regime. Without the embedding the same network returns straight lines.



Running it backwards

Freeing θ costs accuracy through gradient starvation and autodiff derivative bias. A derivative-free integral residual removes the second cause:

$$r_k = (z_{k+1} - z_k) - \frac{\Delta\tau_k}{2} (f_k + f_{k+1})$$

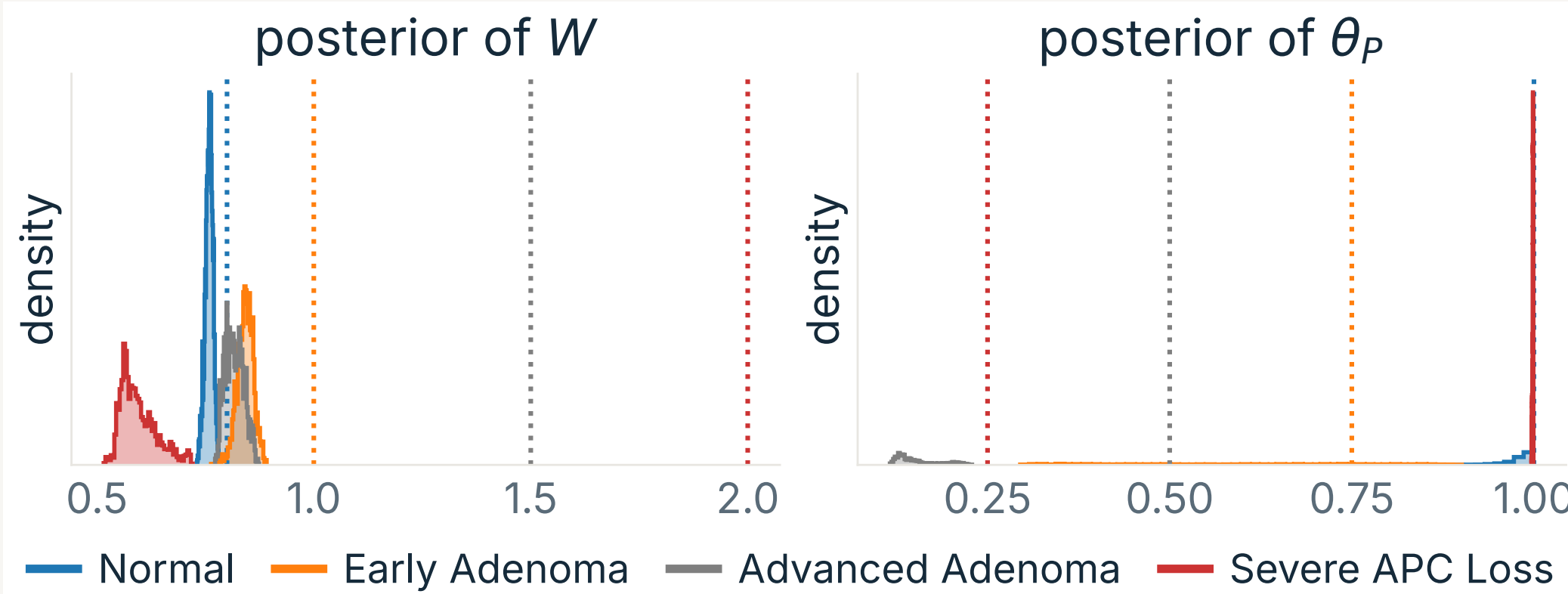


True against recovered, for three of the best-converging parameters.

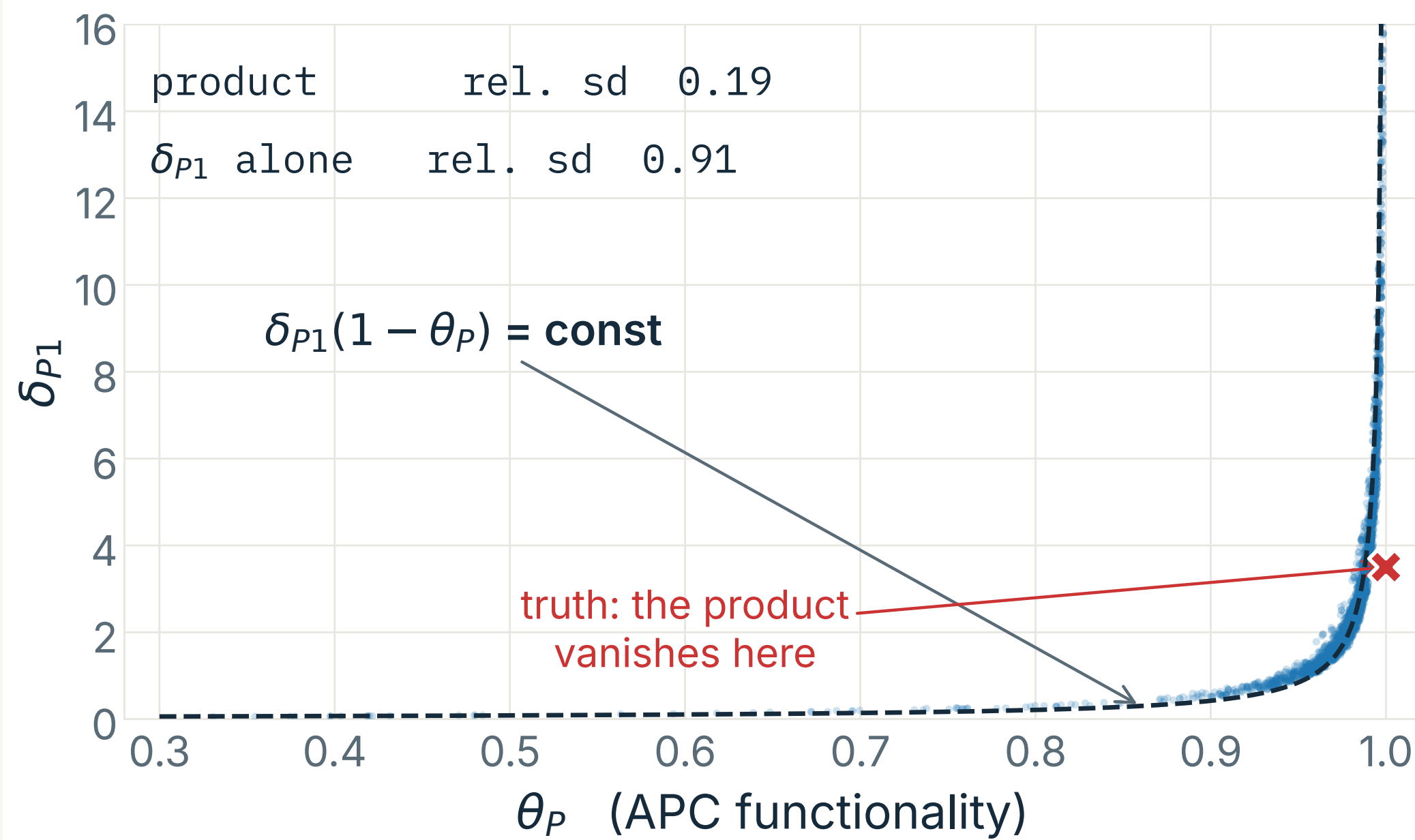
Parameters recovered under 10% error, four regimes: **10/4/5/7** $\rightarrow$ **17/16/10/7**, a total of **37** $\rightarrow$ **50** of 144. Severe APC Loss does not move. What is left there is structural.

3 A posteriori says what data cannot pin down.

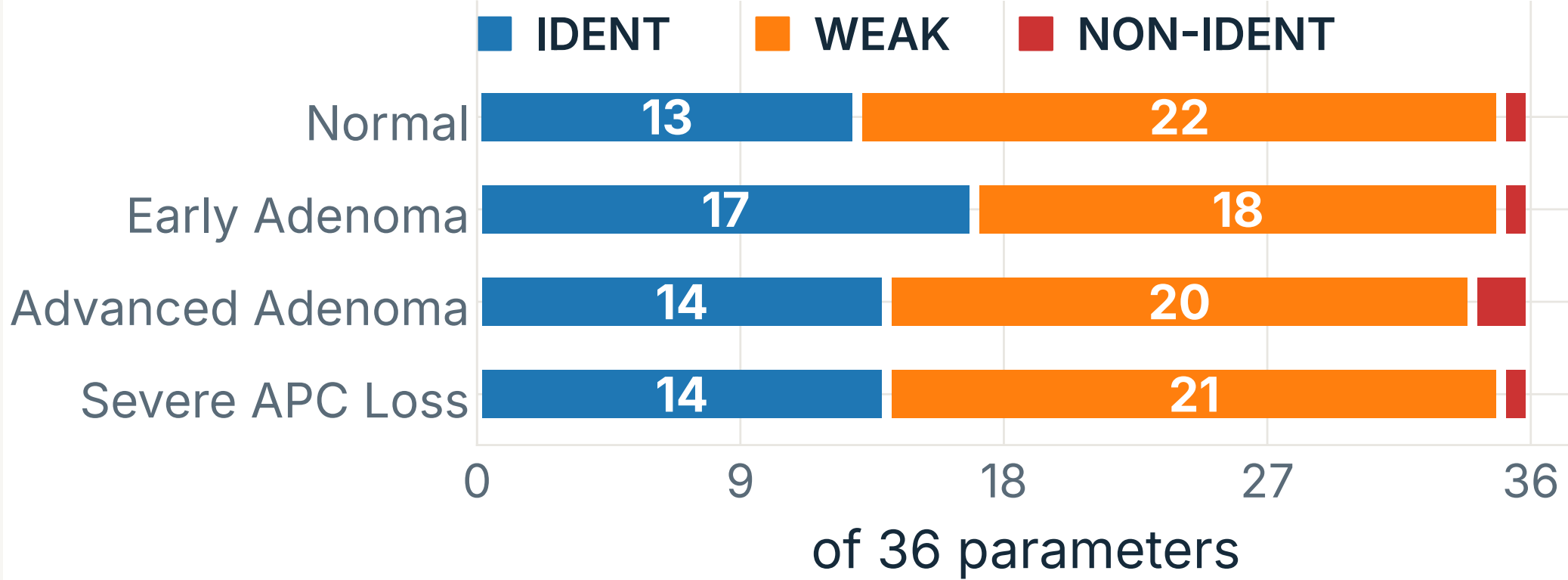
HMC over the 36 parameters with the state networks frozen, derivative-free because the likelihood reuses the integral residual.



W stays tight in every regime. θ_P reverts toward its prior as the WNT drive rises. That is the high-WNT wall, stated as a posterior width.

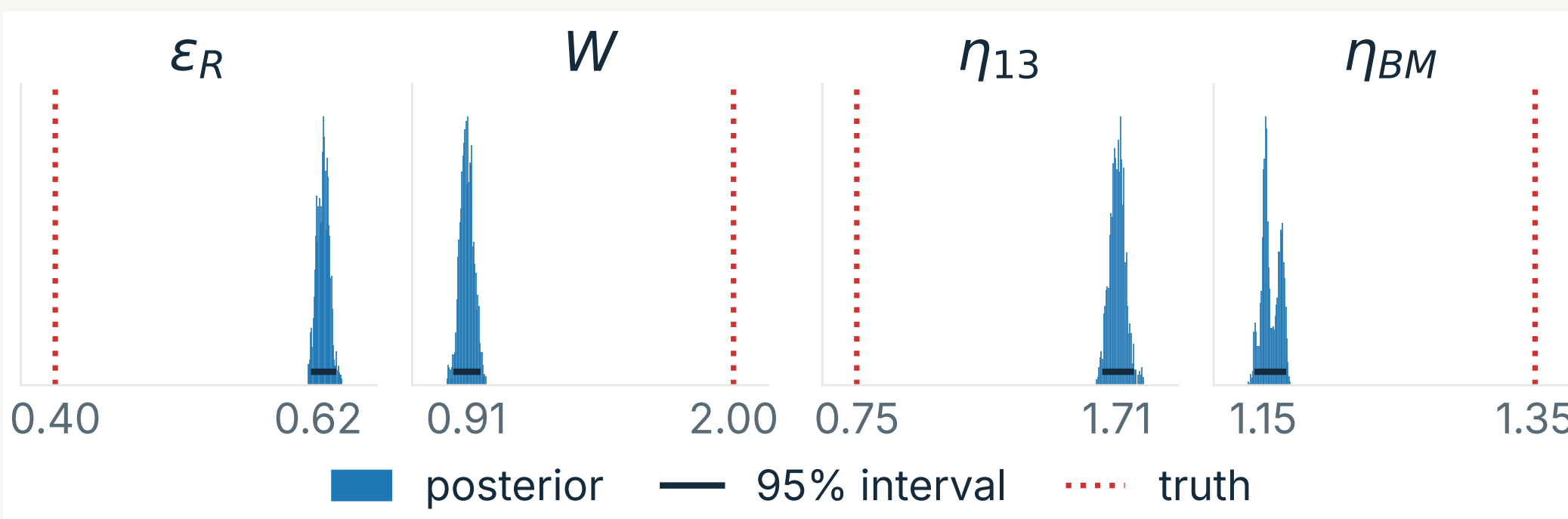


The sampler rides a curved valley unprompted. The product $\delta_{P1}(1 - \theta_P)$ is $4.7\times$ better constrained than δ_{P1} alone, and at the healthy truth $\theta_P = 1$ it vanishes, so δ_{P1} is free.



The Fisher information agrees from a third direction, degrading with WNT drive.

The first run said 36 of 36. It was an artifact.
The physics likelihood summed 210k residual terms against a tiny σ , giving a pinpoint, biased posterior. Honest recovery is 19/20/11/8, and the run still fails its own sample-size and coverage gates. We claim the geometry and the diagnosis, never the widths.



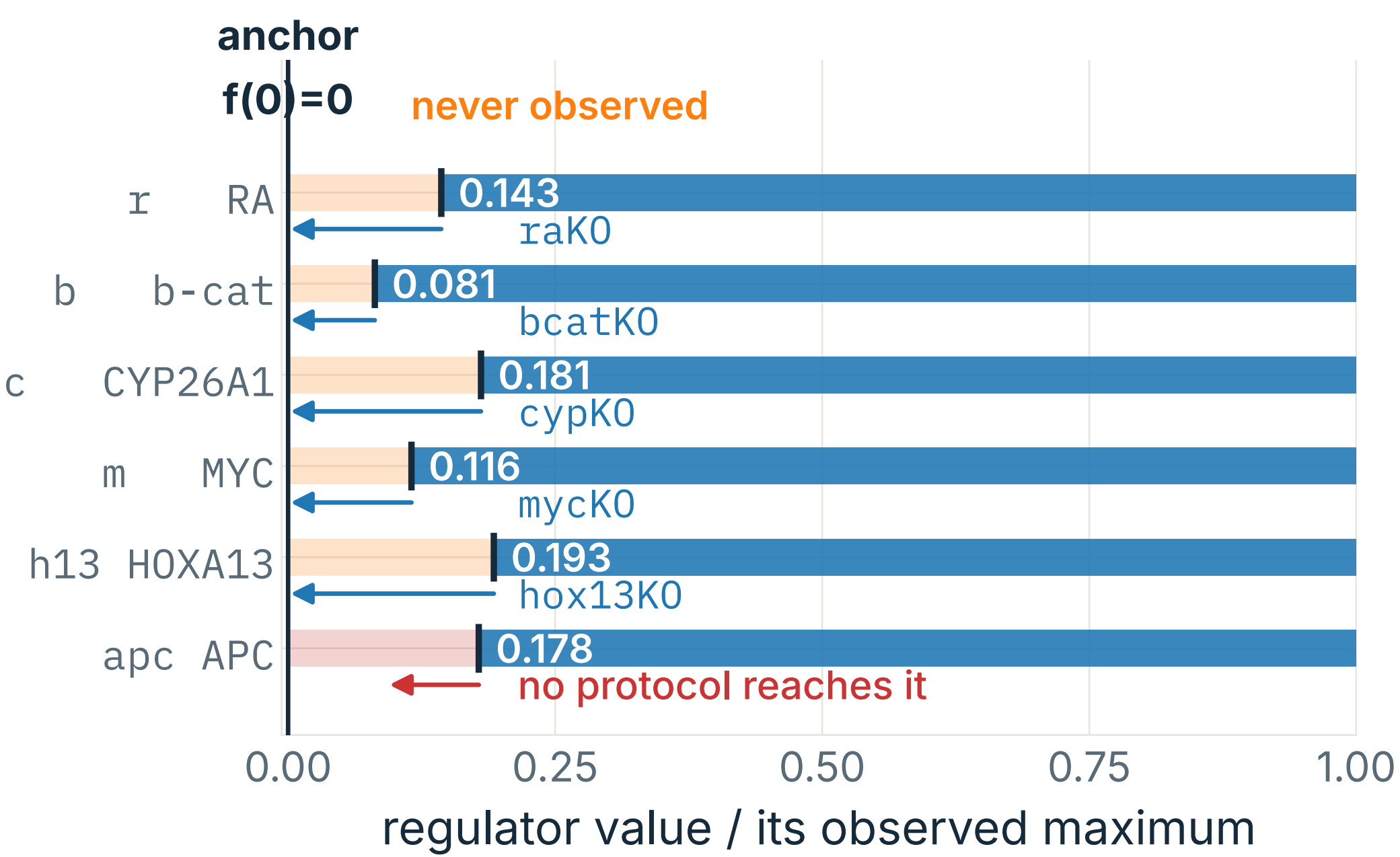
Severe APC Loss, the four worst parameters. Each truth sits outside its own 95% interval, and coverage is 13 of 36.

4 A constraint is only as good as the data that visit it.

$$\varepsilon \dot{x} = a + f_{\text{NN}}(u) - x, \quad f_{\text{NN}}(0) = 0$$

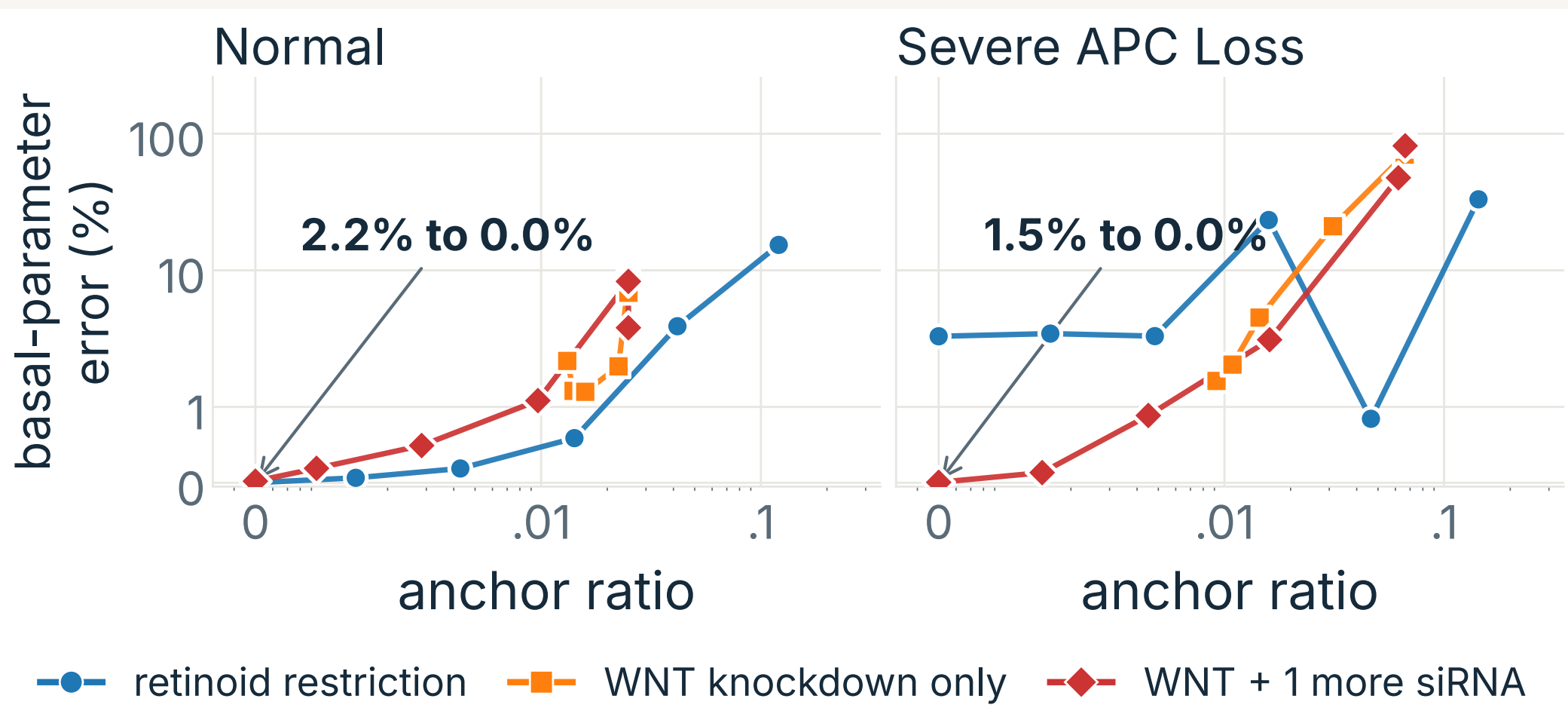
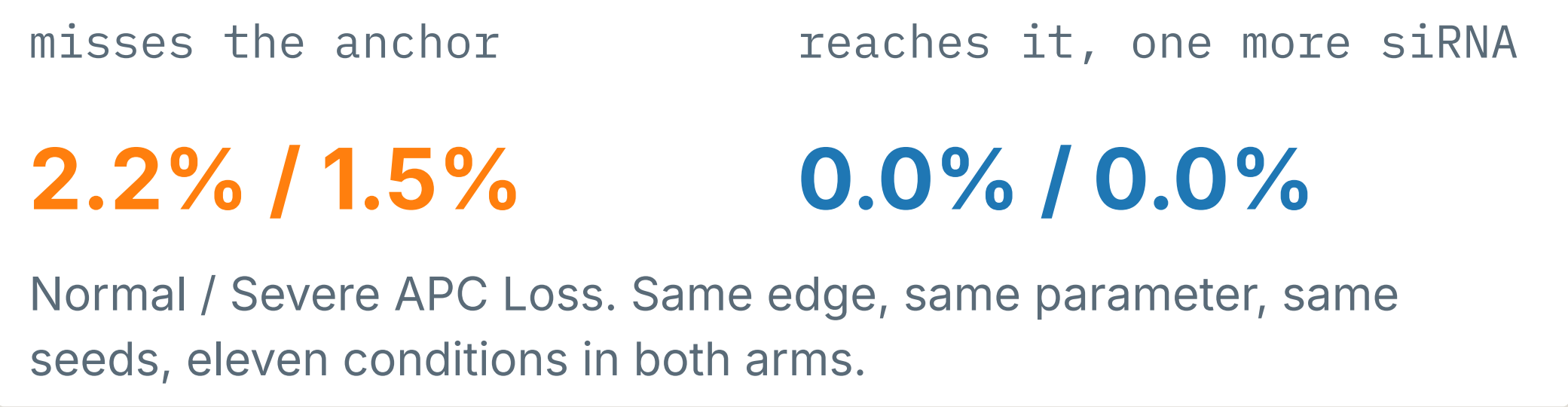
The anchor is meant to stop the network absorbing a constant out of the basal-production parameter a .

It does not. Across five parameterisations, including the monotone and bounded one the literature recommends, the basal parameter comes out **14 to 203%** wrong.



No regulator ever approaches zero, in any of 10 conditions or 4 regimes. The anchor is asserted at a point the data never visit.

The fix is an experiment



Against the anchor ratio the arms collapse onto one curve. Against dose they do not. The anchor is the governing variable.

Not shown: an information-matched control that gains nothing in 8 of 8 cells, and a prospective test at 95.6% $\rightarrow$ 0.0%. Restart noise is ± 0.5 to 0.9 pp, on a single seed, so there are no error bars.

Takeaways

- 1 Nondimensionalization and Fourier features make a stiff 7-ODE system solvable by a PINN from sparse data.
- 2 Inverse recovery is limited by information, not architecture.
- 3 A constraint on a learned term carries information only where the data have support. The remedy is experiment design, computable in advance.